

Satellite Image Land-Use Classifier & Temporal Change Detector

Transfer learning, embedding-based change detection, and an interactive dashboard over EuroSAT & UC Merced.

1. Problem & data

We classify Sentinel-2 satellite tiles into 10 land-use categories and detect land-cover change between two time periods. Primary dataset: EuroSAT (27,000 RGB tiles, 64x64, 10 classes). Holdout: UC Merced Land Use (2,100 images, 21 classes), used zero-shot via a documented 21->10 class semantic mapping (config.UCMERCED_TO_EUROSAT) to test domain generalization rather than re-training on it.

EuroSAT ships without per-tile geo-coordinates. Because the original extraction walked each Sentinel-2 scene raster-by-raster, file-adjacent tiles are also ground-adjacent, so we treat contiguous runs of 25 same-class tiles as a 'spatial block' and split whole blocks (never individual tiles) 70/15/15 into train/val/test -- see Section 4.

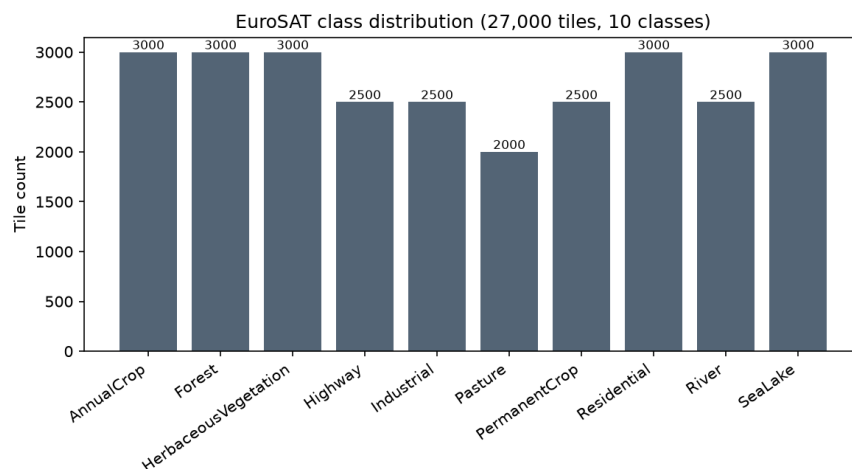


Figure 1. EuroSAT class distribution (27,000 tiles).

2. Module 1 -- Land-Use Classifier

Baseline: a scratch 3-layer CNN (conv-BN-ReLU-pool x3 + 2-layer head), no pretraining. Transfer model: ImageNet-pretrained ResNet-18, two-phase fine-tuning -- phase 1 freezes everything but the classifier head (3 epochs); phase 2 unfreezes layer3+layer4 at LR/10 for 5 more epochs. ResNet-18 was chosen over EfficientNet-B0 because its 512-dim penultimate feature vector is exactly the embedding size Module 2 requires.

Model	Macro-F1 (EuroSAT val)
Baseline scratch CNN	0.853
ResNet-18 (two-phase)	0.970
ResNet-18 -> UC Merced holdout (zero-shot)	0.291

Frozen vs. unfrozen ablation (equal 8-epoch budget):

Configuration	Val accuracy
Frozen backbone for all 8 epochs	0.890
Two-phase, after phase 1 (3 epochs)	0.877
Two-phase, final (+5 unfrozen epochs)	0.971

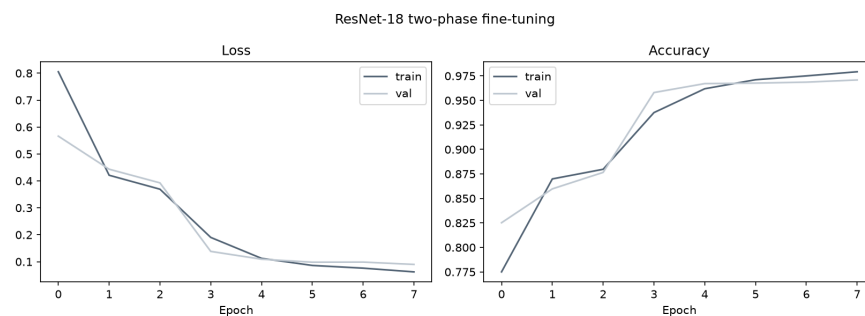


Figure 2. ResNet-18 two-phase fine-tuning loss/accuracy curves.

3. Confusion matrices

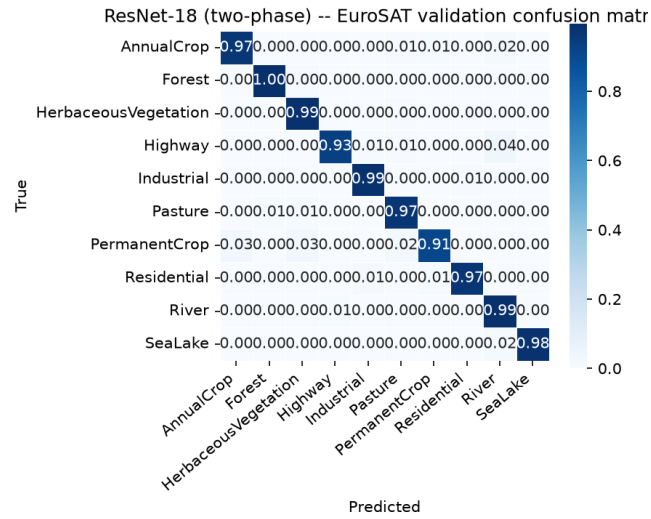


Figure 3a. ResNet-18 on EuroSAT validation.

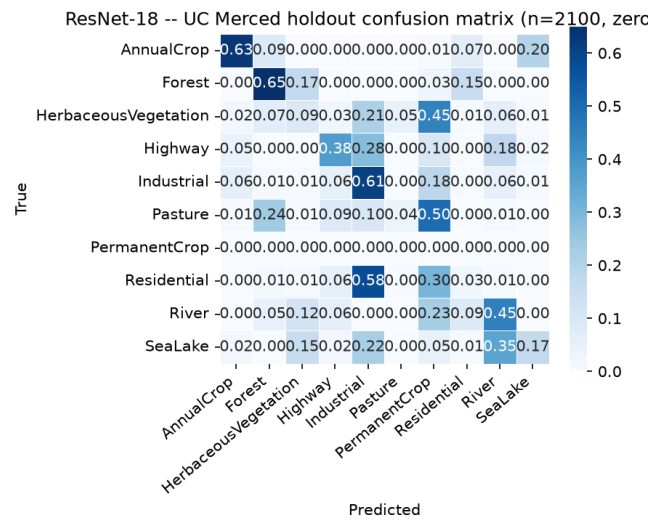


Figure 3b. ResNet-18 on UC Merced holdout (zero-shot, class-mapped). PermanentCrop has no confident UC Merced analogue and is intentionally left with zero support.

4. Spatial leakage experiment

Split strategy	Val accuracy
Spatial block split (honest)	0.859
Random per-image split (leaky)	0.874
Leakage gap	+0.015

EuroSAT tiles were extracted by walking each Sentinel-2 scene raster-by-raster, so tiles with adjacent file indices are also adjacent on the ground and visually near-duplicate (same field, same cloud shadow, same crop row pattern). A random per-image split scatters these near-duplicates across train AND val, so the model partly memorizes local texture instead of learning a generalizable class boundary -- inflating val accuracy. The spatial-block split keeps every tile from one ~25-tile contiguous block on the same side of the split, removing that leakage channel. The gap above is the cost of that leakage.

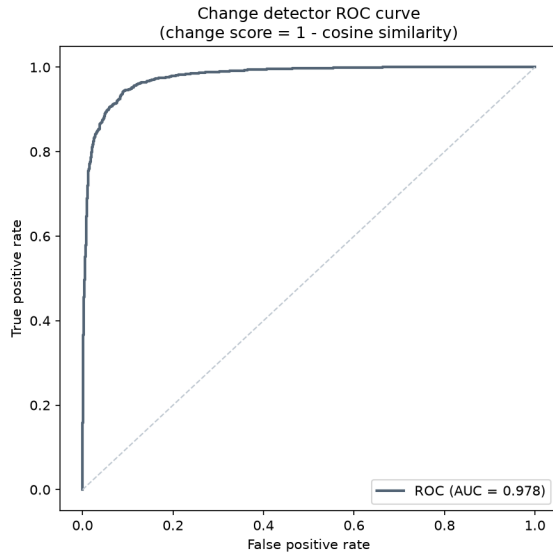
Bonus D -- class imbalance experiment

Downsampled Pasture, PermanentCrop to 20% of their training size, then compared per-class F1 with vs. without class-weighted cross-entropy loss.

Class	No mitigation F1	Weighted-loss F1
Pasture	0.690	0.747
PermanentCrop	0.037	0.676

5. Module 2 -- Temporal Change Detector

The fine-tuned backbone (head stripped) extracts a 512-dim global embedding per tile. EuroSAT has no real before/after pairs, so a synthetic T1/T2 series is built from the spatial blocks: 'unchanged' pairs draw two tiles from the same block (label 0); 'changed' pairs draw tiles from two different-class blocks (label 1). Because ground truth is known by construction, a real ROC curve can be computed for the cosine-similarity change score.

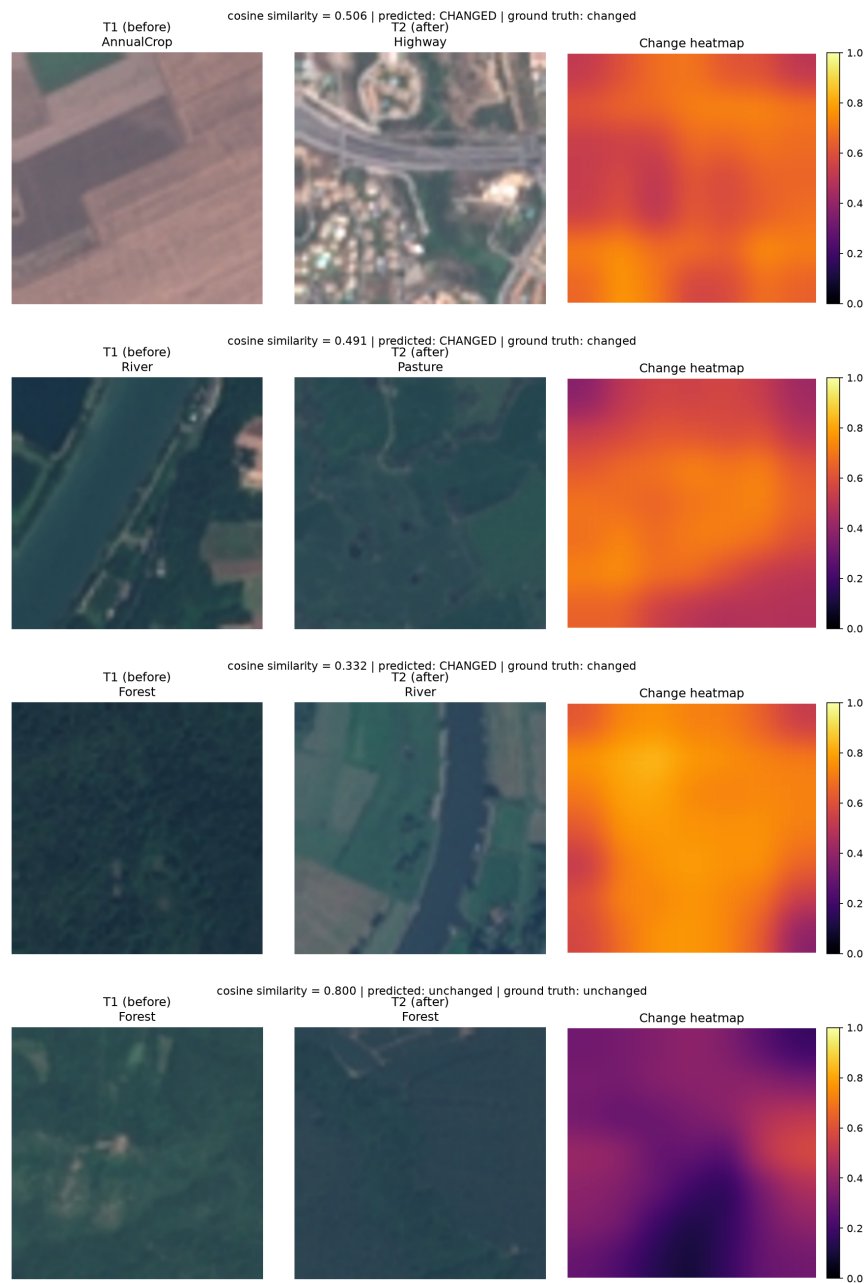


Operating point	Sim. threshold	TPR	FPR
high recall	0.661	0.95	0.11
balanced	0.651	0.94	0.09
high precision	0.605	0.89	0.05

Figure 4. ROC curve (AUC=0.978) and the three bonus-B operating points (high recall / balanced via Youden's J / high precision), toggleable in both the Streamlit dashboard and the Next.js change-detection page.

6. Sample change heatmaps

Each tile's 7x7 spatial feature map (pre-pool layer4 output) is compared patch-by-patch via cosine distance, upsampled, and overlaid -- showing *where* in the tile the embedding shifted, not just a single global score.



7. Error analysis



Figure 5. Top-5 most confidently wrong predictions, EuroSAT validation split.

SeaLake → **River** (conf=1.00): Both are water; the model may be keying on hue/turbidity rather than the shape cues (linear vs basin) a human would use.

Highway → **River** (conf=1.00): Both render as long, thin, gently-curving linear features bisecting the tile; asphalt-grey and water-grey are easy to confuse under haze or shadow.

Pasture → **HerbaceousVegetation** (conf=1.00): Both are green, low-texture, non-forested ground cover with no strong distinguishing edges at 64x64 resolution.

Highway → **River** (conf=1.00): Both render as long, thin, gently-curving linear features bisecting the tile; asphalt-grey and water-grey are easy to confuse under haze or shadow.

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Bonus A/C -- GradCAM & embedding visualisation

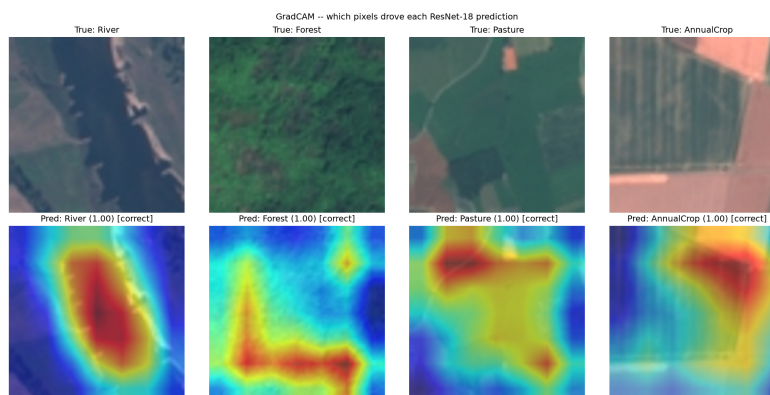


Figure 6. GradCAM overlays -- which pixels drove each prediction.

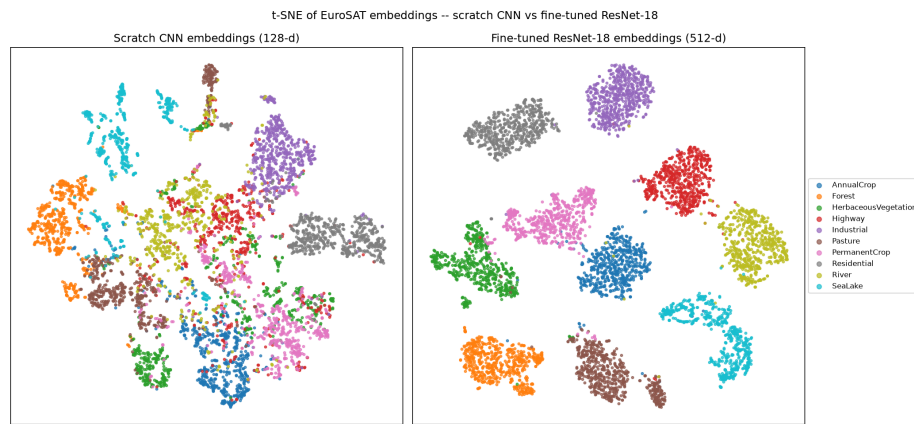


Figure 7. t-SNE of 6,000 EuroSAT embeddings -- scratch CNN vs. fine-tuned ResNet-18.

8. Limitations & conclusion

(1) The T1/T2 change series is synthetic -- EuroSAT has no real repeat acquisitions, so the ROC curve measures the detector's ability to separate same-block from different-class-block pairs, not real seasonal/sensor drift. (2) The UC Merced holdout uses a manually authored, debatable 21->10 class mapping; one EuroSAT class (PermanentCrop) has no mapped UC Merced analogue and is untested. (3) Tiles are evaluated independently; no temporal smoothing or multi-tile spatial context is used. (4) The spatial-block split approximates geography via file-index adjacency, not real lat/lon.

Despite these constraints, the two-phase fine-tuned ResNet-18 substantially beats the scratch baseline and generalizes non-trivially to a different sensor/dataset (UC Merced), and the cosine-similarity change detector achieves a strong, threshold-justified ROC AUC on the constructed change task.